

VAT Refund BI Dashboard — Uzbekistan

MSc Business Intelligence & Analytics · Westminster International University in Tashkent

Student ID: 002110 · Period: 2020–2024

Live · Interactive · What-if Analysis

Year

All years

Sector

All sectors

Metric

Cash Flow

n = 200 observations

AVG OPERATING CASH FLOW

-446

mln UZS

AVG REFUND DELAY

62.9

calendar days

AVG CURRENT RATIO

1.35

target ≥ 1.5

REFUND COVERAGE RATIO

75.9%

2024 (up from 28.7% in 2020)

Cash flow vs refund delay

Scatter by sector · $\beta = -10.39$ · $r = -0.723$ · $p < 0.001$

Average cash flow by sector

Mean operating cash flow (mln UZS)



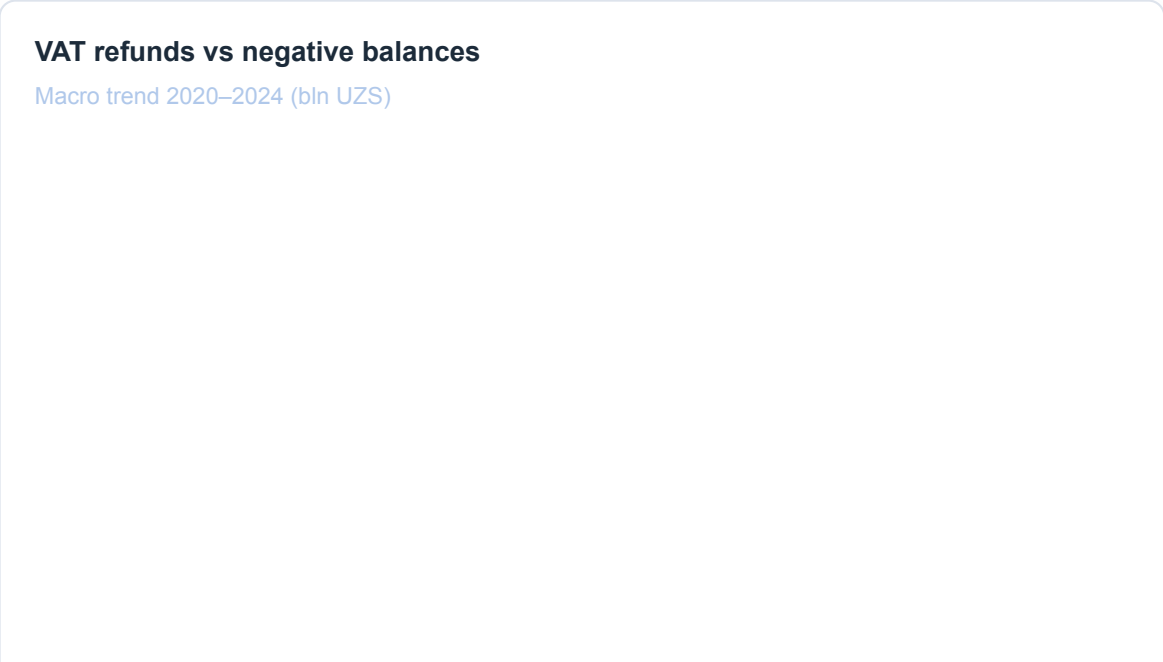
Refund delay by sector and year

Average processing days



VAT refunds vs negative balances

Macro trend 2020–2024 (bln UZS)





What-if Analysis — Liquidity Stress Test

Decision Support

Simulate how changes in VAT refund processing time affect enterprise liquidity. Move the sliders to model different scenarios.

VAT Refund Processing Time

62 days



15 days (best)

120 days (worst)

Annual Turnover

4 800 mln UZS



500 (small)

15,000 (large)

Sector

Services (lowest risk)



● AMBER ZONE — MODERATE RISK

Resulted Cash Flow

60.7 mln UZS

Predicted Cash Flow	-89.7 mln UZS
Predicted Current Ratio	1.36
Quick Ratio (est.)	1.11
Frozen VAT / Cash Flow	18.3%
Risk Score	1.00

Regression formula applied:

Cash Flow = 207.34 – 10.39 × Days + 0.031 × Turnover

Risk classification:

- Green: delay < 45 days, CF > 0
- Amber: delay 45–75 days, CF –100 to 0
- Red: delay > 75 days, CF < –100 mln

Policy insight:

Each +10 days delay → –103.9 mln UZS cash flow impact

Lag effect (Section 4.6.4):

Maximum liquidity shock occurs at t+1:

$$\beta(t) = -10.39 \cdot \beta(t-1) = -12.14 \cdot \beta(t-2) = -8.73$$

Source: State Tax Committee of the Republic of Uzbekistan & anonymised enterprise dataset (E001–E050, n=200) | $r = -0.723$ · $R^2 = 0.523$ · $\beta = -10.39$ · $p < 0.001$ | MSc BIA Dissertation · WIUT 2024–2025